



GigaCloud

Installation, Upgrade and Removal Guide

July 2019

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Introduction

This guide will assist you in properly configuring prerequisites, initial installation of GigaCloud (GTEP), subsequent upgrades and removal of the software.

GigaCloud provides for installation in the following configurations:

-  Standalone, Single GTEP Server Instance
-  Clustered GTEP Instance Configuration

Overview

Microsoft Active Directory Rights Management Services (AD RMS) must be installed and working properly before GigaTrust Enterprises Plus can be installed and configured.

GigaCloud Administration is installed on the AD RMS server(s).

The GTEP installation delivers some or all of these items depending on the particular installation:

- **GTEP Services**

1. GTEP Log Service
2. GTEP Router Service
3. GTEP Server

- **Web Components**

4. GTEP Administrator (Default GigaCloud Administration UI)
5. Legacy GTEP Administrator (Deprecated)
6. Legacy Hosting Administrator (Deprecated)
7. Web Services
8. HTTP Modules
9. REST Module (Required If Using The AD RMS Mobile Device Extension)
10. External Web Client (Required If Using Managed Users)
11. Front Page Web (GigaCloud Trial Website)

- **GTEP Tools:**

12. GTEP Configuration Tool (Includes the GTEP DB Setup tool)
13. Certificate Admin Tool
14. GTEP PowerShell Module
15. GTEP Monitor (monitors GTEP application components and restarts them if they stop)

The GTEP Installer also does the following:

- Sets up a GTEP SQL database.
- Adds GTEP Managed Modules for Authentication in IIS to the AD RMS certification and licensing pipelines.

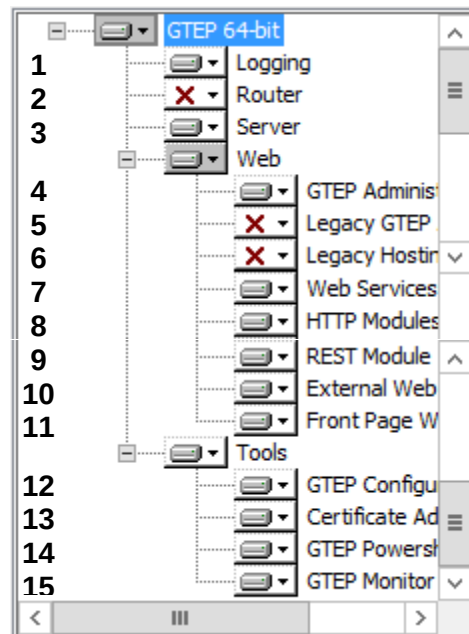


Figure 1 - Installation Options in the GTEP Installer (Tree fully expanded for clarity)

- Creates a virtual directory under the AD RMS _wmcs virtual directory in IIS for GigaTrust that contains web services for the GigaTrust Client and Policy Agent.
- Creates certificates required for the GTEP application.

Installing GigaCloud involves preparing the Windows Server environment for GTEP, then installing and configuring the GTEP software.

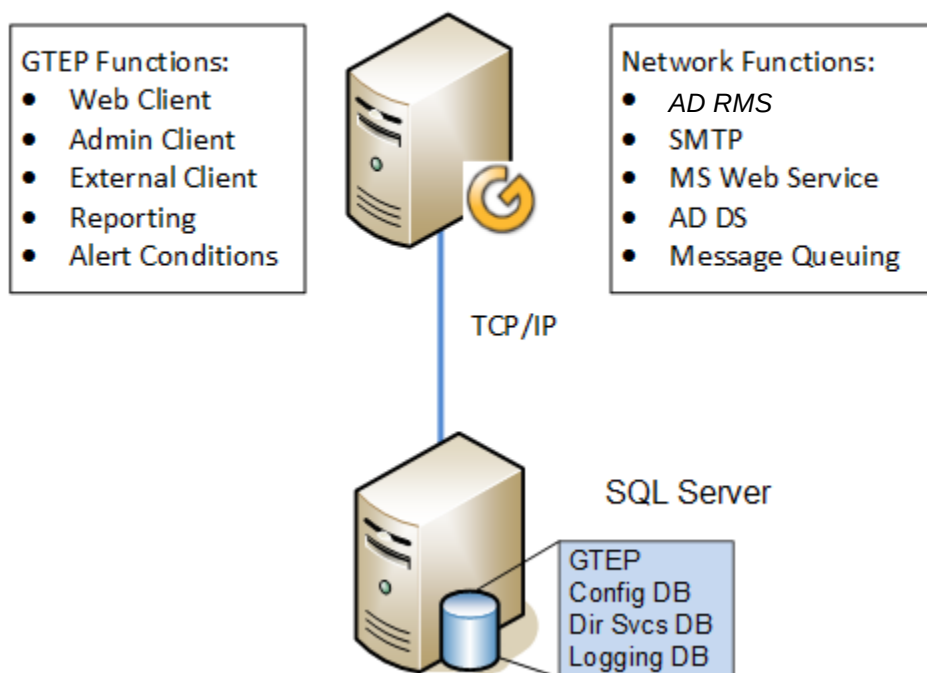
GigaCloud Configurations

Standalone GigaCloud Instance

GTEP is powered by a Windows Communication Foundation (WCF) application server running as a system service, named GTEP Service. WCF is Microsoft's most recent communications system. It is a robust, fast, and reliable framework for inter-process communication both on a single machine and across multiple machines.

GTEP Service runs as the AD RMS Service User and communicates with other resources on the network, including databases, an SMTP server and other Web Services, Web Applications, and WinForms clients.

Standalone GTEP-AD RMS Server

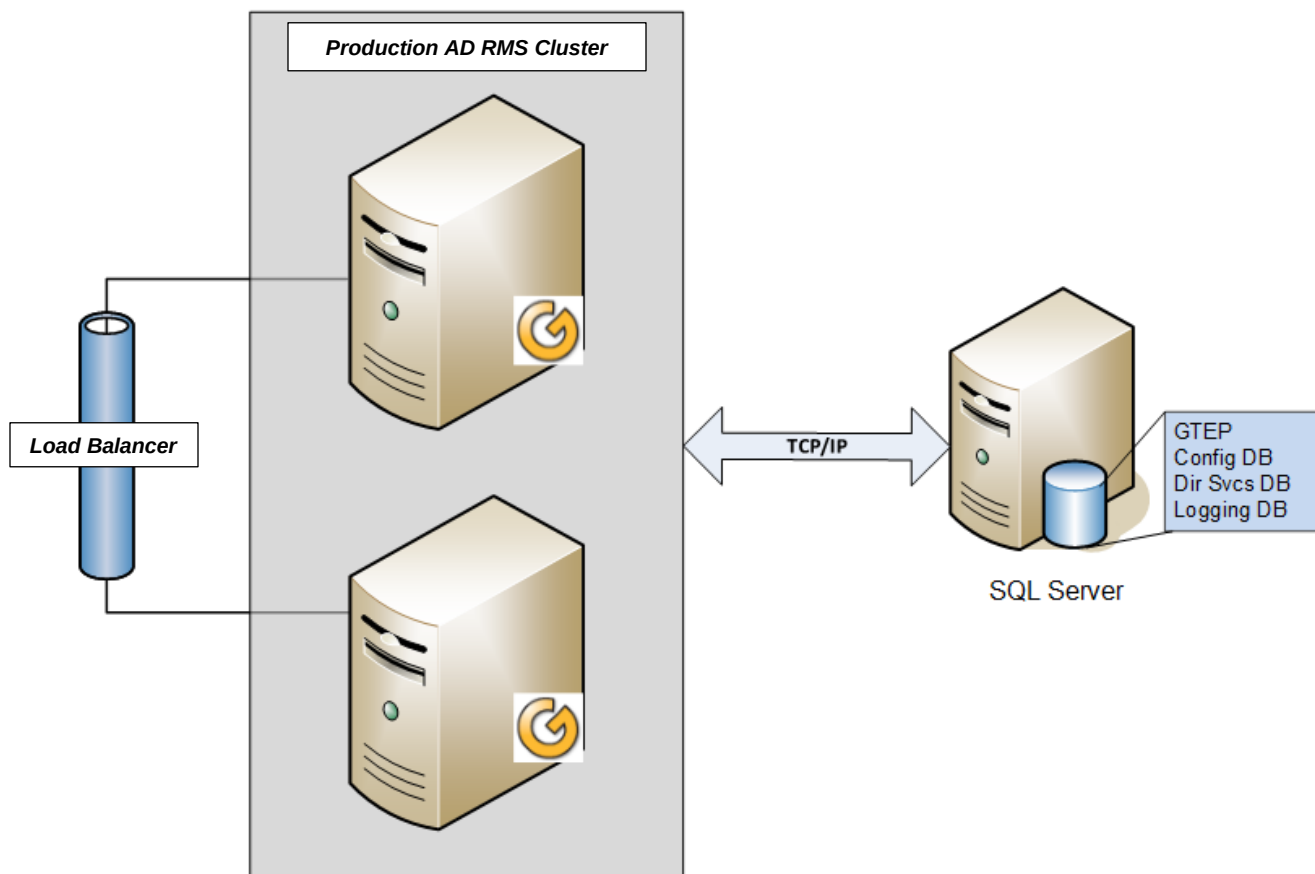


For Standalone, Single instance GTEP Configuration, continue with this guide below...

GigaCloud Clustered Instance

GTEP Servers can be configured to run in a parallel cluster, similar to an RMS cluster. In this configuration, GTEP Servers are replicas of one another and can be load balanced and they share a single database. Parallel servers will appear in the GTEP Administration Sitemap as a single entry.

GTEP Clustered Instance



Load Balancing and Cookie Persistence

If the configuration is a clustered instance using a load balancer, the `ASP.NET_SessionID` cookie must be persisted in the configuration.

Note: GigaTrust does not provide support for load balanced configurations other than to confirm a specific setting is required.



For Clustered GigaCloud Configuration, start with the [GigaCloud Clustered Configuration Guide](#).

Prerequisites

The following configuration steps are required before GigaCloud Administration can be installed. If you do not have access rights to the AD RMS Service Account, you can create a GTEP Service Account to accomplish the same permissions prerequisites. See [Appendix D](#) below.

1. AD RMS Service Account and Password:

Account: _____

Password: _____

2. Create a Super User Group as a Universal Security AD Group for GTEP Administrators:

Users of this group will have administrative access to the GTEP server. This group does not need to be email-enabled. (Suggested name: **GTEnterpriseAdmins**)

3. Add Users to the GTEP Administrators Group:

1. Add your AD RMS Service account user to the GTEP Administrators group.
2. Add any other users who will need administrative access to GTEP to the GTEP Administrators group.

4. Microsoft .NET Framework 4.6.2 is installed on the GTEP server:



After installing the .NET Framework package, it is recommended that any updates also be installed.

.NET Framework 4.6.2 is required!

Installing GigaCloud



Installation Permissions

To install GTEP, log on to the server with an **account that has administrator rights** to install and write to the registry.

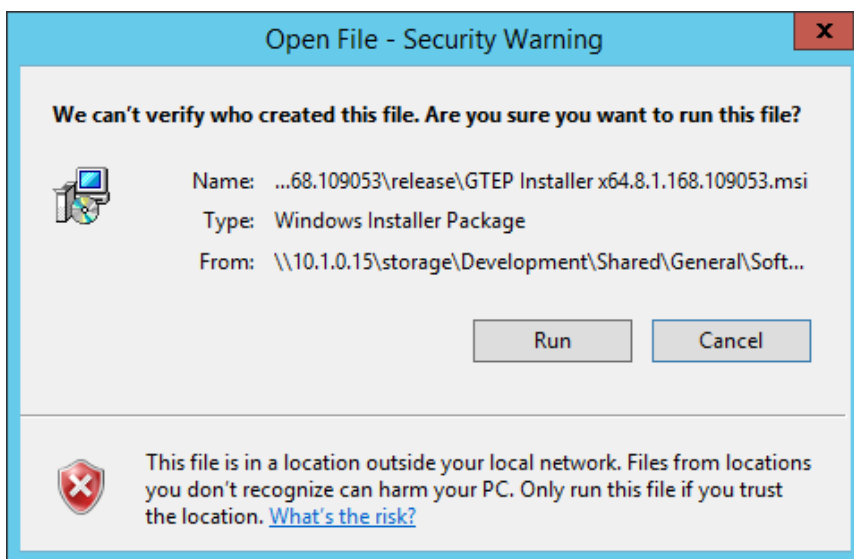
To create the GTEP database during the installation, the logged on User account must be a member of the following **SQL Server roles: sysadmin**.

Note: The sysadmin role is required if replacing an existing GTEP database.

To modify the AD RMS configuration and logging databases during the installation, the log on User account must have db_owner rights on those databases. The installation adds stored procedures to the AD RMS configuration database and provides appropriate rights to the configuration and logging database for GTEP to properly function.

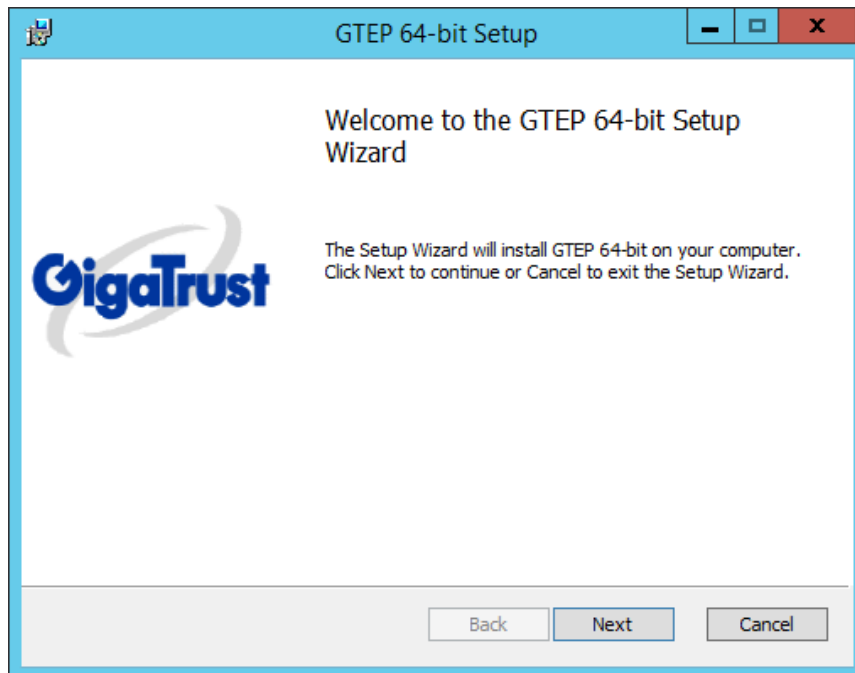
1. Start the Installation:

Run **GTEP64VERSION.MSI**.



2. The Welcome screen appears.

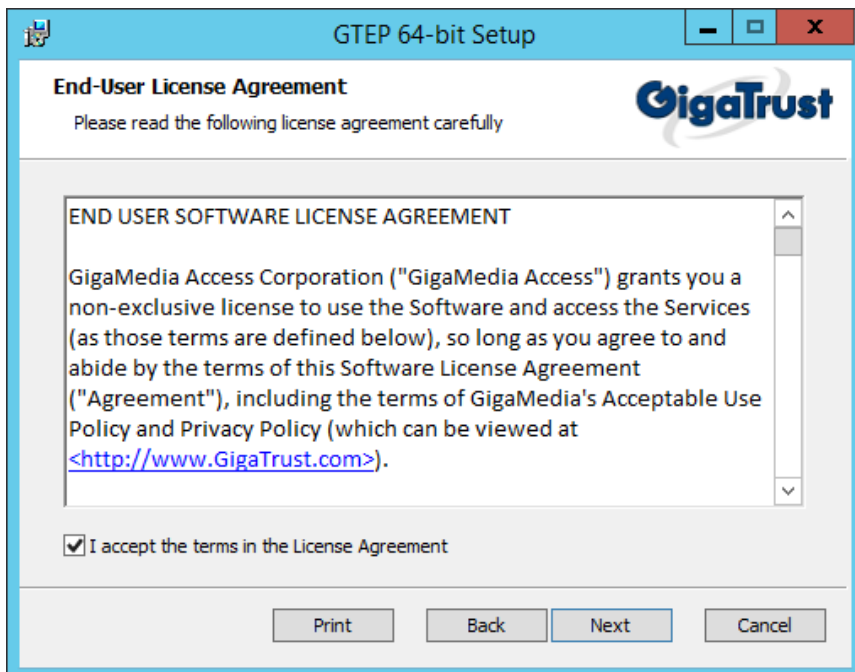
Click **Next** to continue.



3. **End-User License Agreement:**

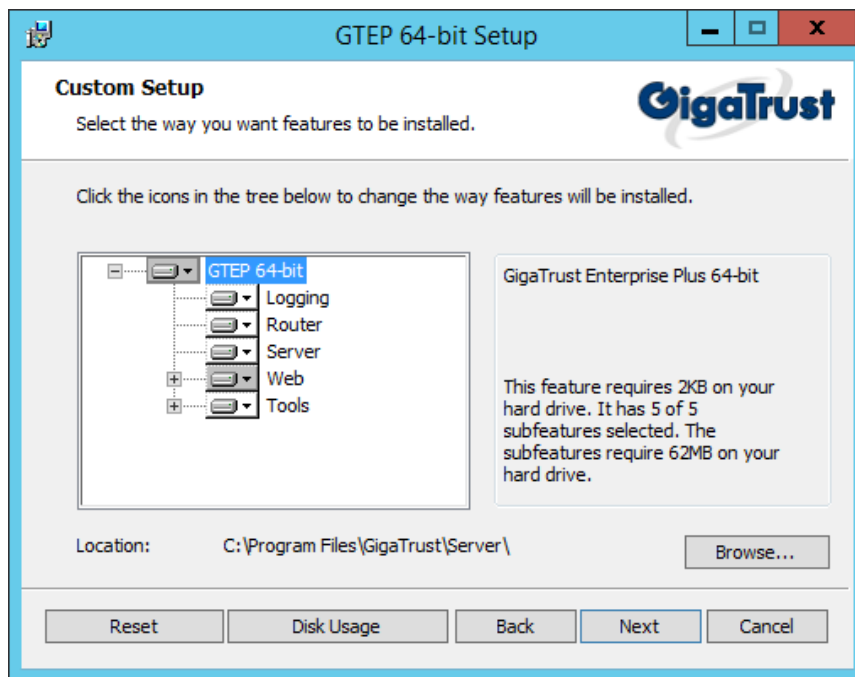
Read and accept the **GigaTrust End-User License Agreement**.

Click **Next** to continue.



4. Custom Setup:

Install with the default features for standalone, single instance GTEP configuration. If you are reinstalling from backup, the previously installed features will already be selected.

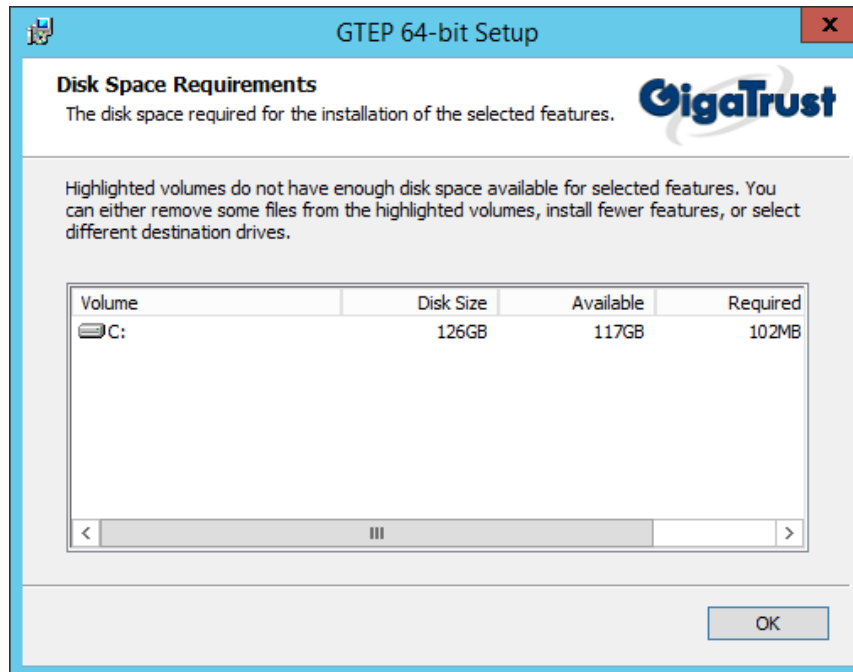


The following are optional features:

1. **Logging** – This component is pre-selected, but optional. It is recommended that you install it to assist you with troubleshooting GTEP installation and operation.
2. **Router** – Not required.
3. **Web > Legacy GTEP Administrator (Deprecated)** – This is the original Administration UI.
4. **Web > Legacy Hosting Administrator (Deprecated)** – This is the original Hosting UI.
5. **Web > External** - This component is required if you need to configure and administer Managed (External) Users.

The bottom of the screen gives you the following options:

- **Location** – Click on **Browse** if you want to install on another server drive. The default drive is C drive.
- **Reset** – Select to revert back to the default screen settings.
- **Disk Usage** - If you click on Disk Usage, the following window will appear. Click **OK** when finished.

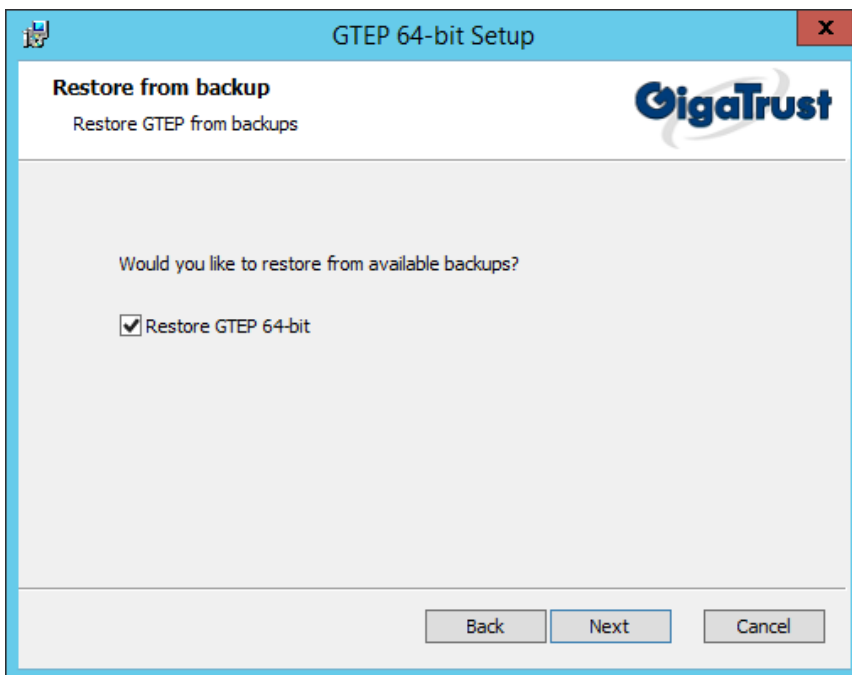


- **Cancel** – Click to cancel and rollback installation.

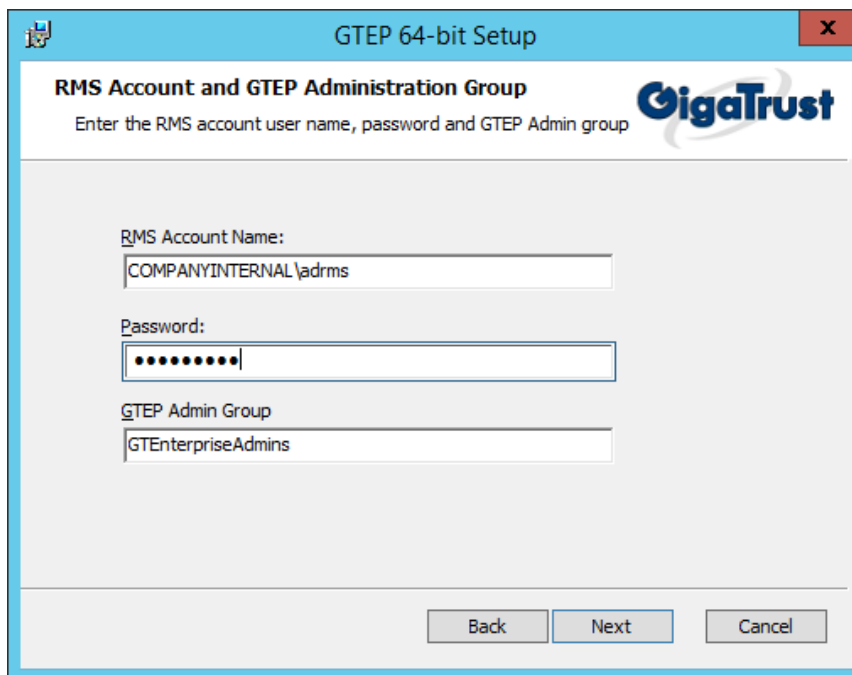
Once you have selected and configured features and options for installation, click **Next** to continue.

5. Restore from backup:

If you are reinstalling GTEP keep this box checked.



Click **Next** to continue.

4. RMS Account and GTEP Administration Group

RMS Account Name: Provide the AD RMS Service Account Name.

Password: Enter the AD RMS Service Account's Password.

GTEP Admin Group: If not prepopulated, enter the name of the GTEP administrators group defined as a prerequisite. Suggested name was **GTEnterpriseAdmins** - Refer to [Prerequisites](#).

Click **Next** to continue.

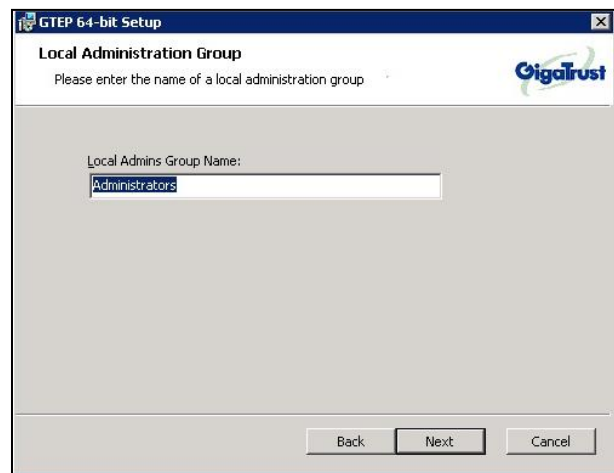
5. Local Administration Group:

Note: This install page is only seen if the local Administration group is not called "Administrators".

Local Admins Group Name:

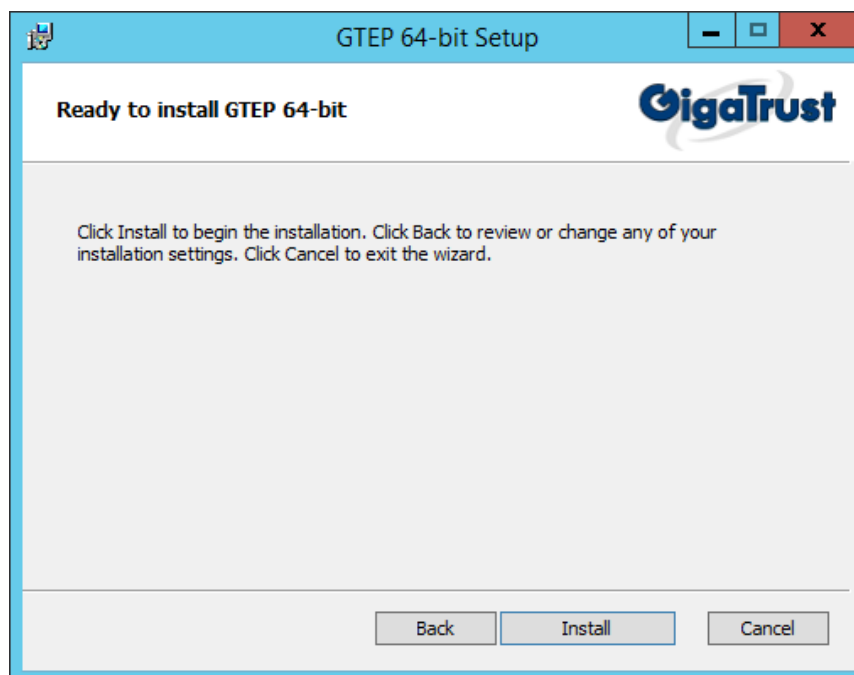
Enter the name of the local administration group

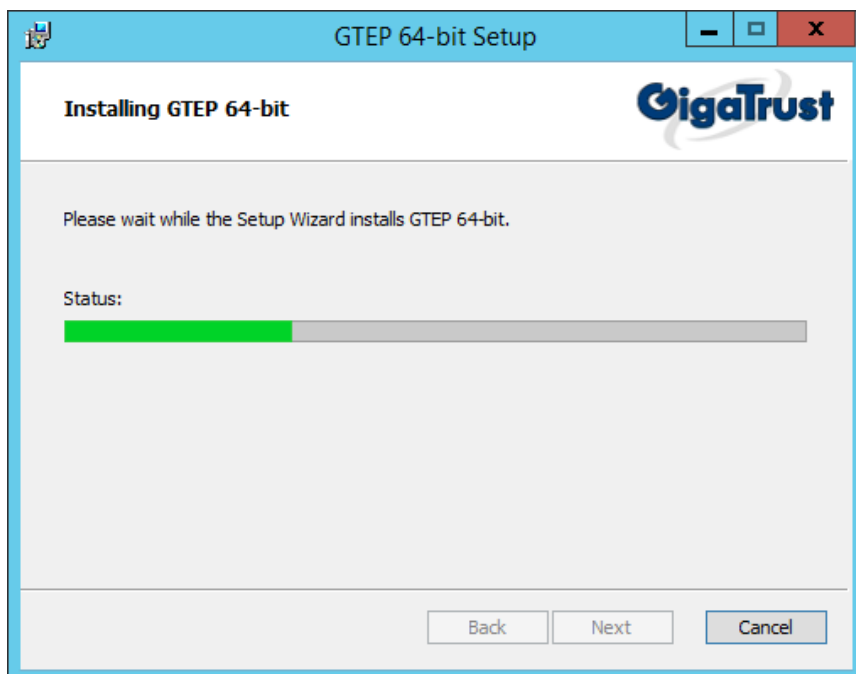
Click **Next** to continue.



6. Install GTEP:

Click **Install** to begin the installation. **Note:** There may be a delay, please click only once.





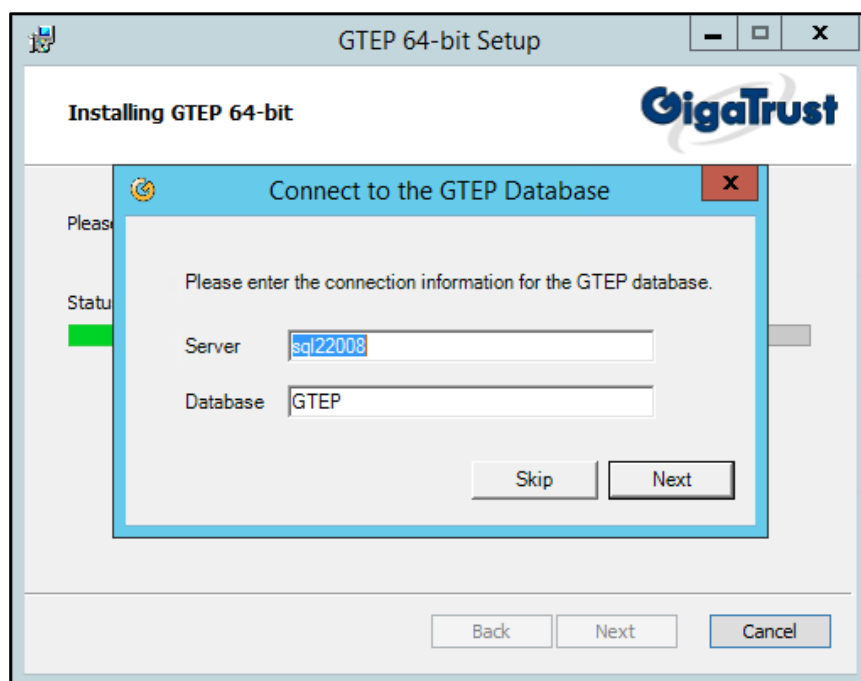
7. Connect to the GTEP Database location:



You have the option to skip the creation of the GTEP database. If **Skip** is selected, run the GTEP DB Setup Utility after the installation is complete (See [Appendix](#) for Reference)



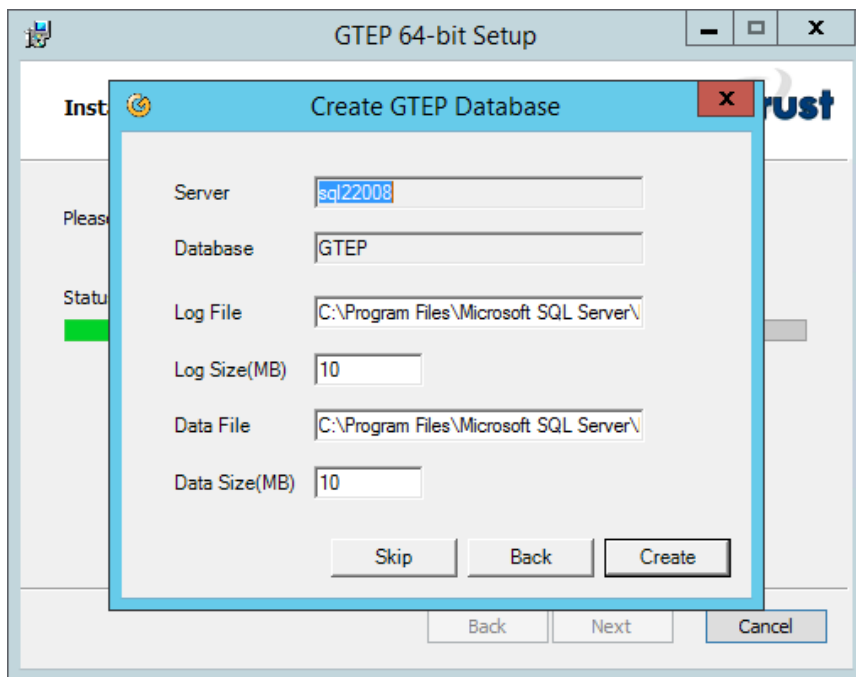
If the installation will become a member of a GTEP cluster, enter the GTEP database connection information used for the previous GTEP server.



Server: Enter the SQL Server instance that will contain the GTEP database. This may be prepopulated.

Database: Enter the name for the GTEP Database. This may be prepopulated.

Click **Next** to continue.



8. Create GTEP Database:

Log File: Enter a valid log file path for the GTEP Database (**Default:** Prepopulated from connection information.)

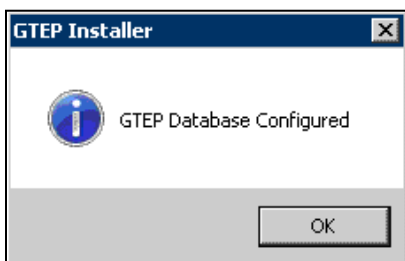
Log Size (MB): Customize the Log File Size file (**Default:** 10MB)

Data File: Enter a valid data file path for the GTEP Database (**Default:** Prepopulated from connection information.)

Data Size (MB): Customize the Data File Size file (**Default:** 10MB)

Click **Create**

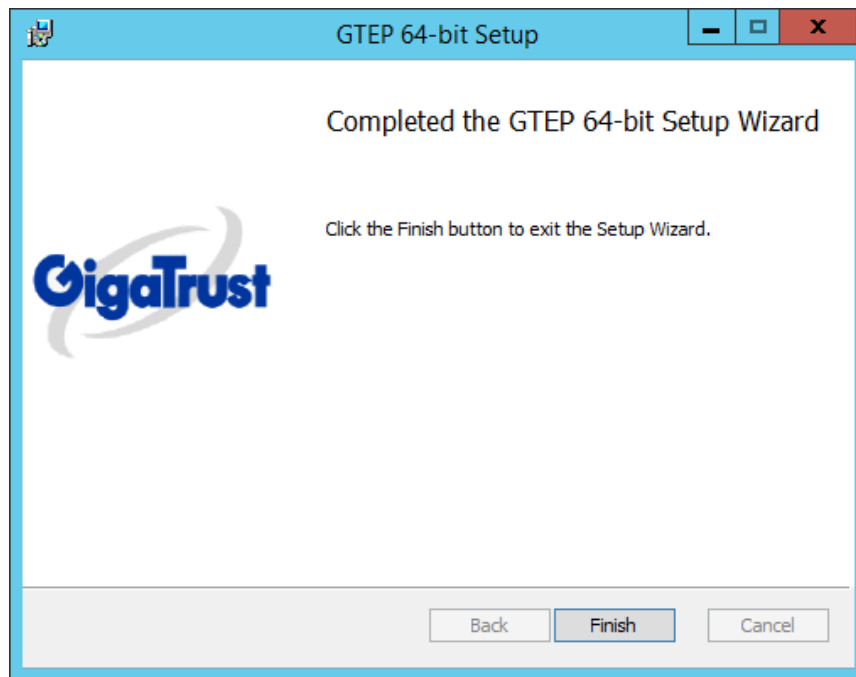
Note: *There may be a delay. Please click only once.*



9. After the GTEP Database is created, click **OK**.

10. Complete installation:

Click **Finish** to complete the GTEP installation.



To finish clustered GTEP configuration, continue with the [GigaCloud Clustered Configuration Guide](#).

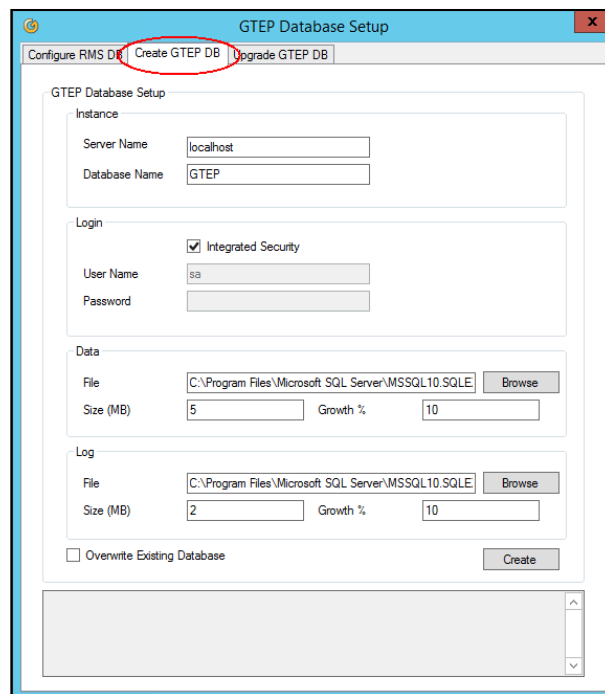
For Single-Instance GTEP continue to the [“Post Installation Configuration for GTEP”](#) section. ➡

SQL Server – Optional Configurations

If Installation Step 7 above was skipped, create GTEP Database.

If you skipped the creation of the GTEP database, you must run the **GTEP DB Setup** tool before proceeding to GTEP Configuration.

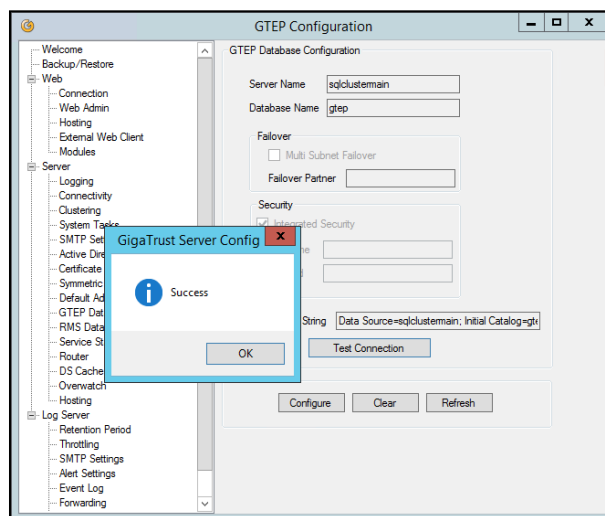
1. **Start** ➤ **All Programs** ➤ **GigaTrust** ➤ **GTEP** ➤ **Tools** ➤ **GTEP DB Setup**
2. Select the **Create GTEP DB** tab.
3. Input the required server information.
4. Click **Create**.



Verify Database Connection

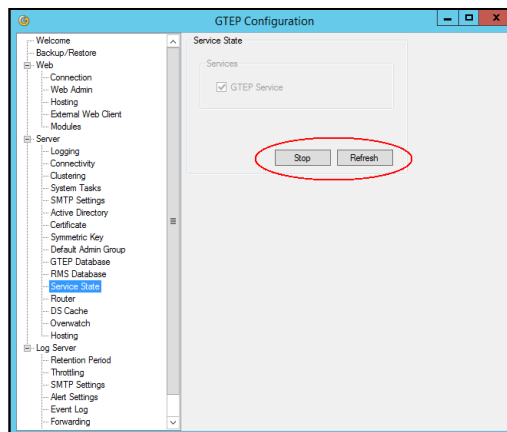
Test the connection to two databases:

1. **Start** ➤ **All Programs** ➤ **GigaTrust** ➤ **GTEP** ➤ **Tools** ➤ **GTEP Configuration Tool**
2. Select the **GTEP Database** node and then click **Test Connection**.
3. Then select the **RMS Database** node and then click **Test Connection**.

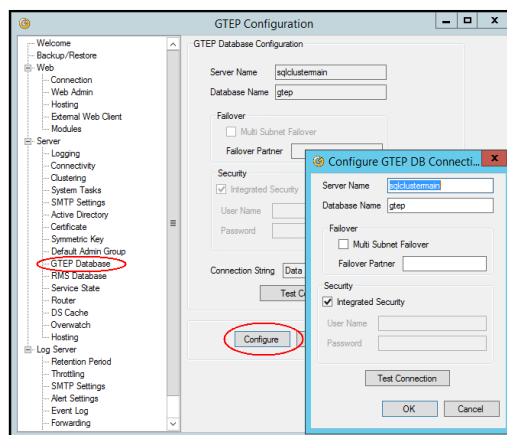


(Optional) SQL Failover Cluster Configuration

1. Start **GTEP Configuration Tool**.
2. In the tree-view in the left-hand pane, select **Service State**.
Click **Stop** to stop the Service State.



3. In the tree-view in the left-hand pane, select **GTEP Database**.
Click **Configure**.
Enter **Failover Partner** and click **OK**.
4. In the tree-view in the left-hand pane, select **RMS Database**.
Click **Configure**.
Enter Failover Partner and click **OK**.
5. In the tree-view in the left-hand pane, select **Service State**.
Click **Start** to start the Service State.



Continue to the “**Post Installation Configuration for GTEP**” section. ➡

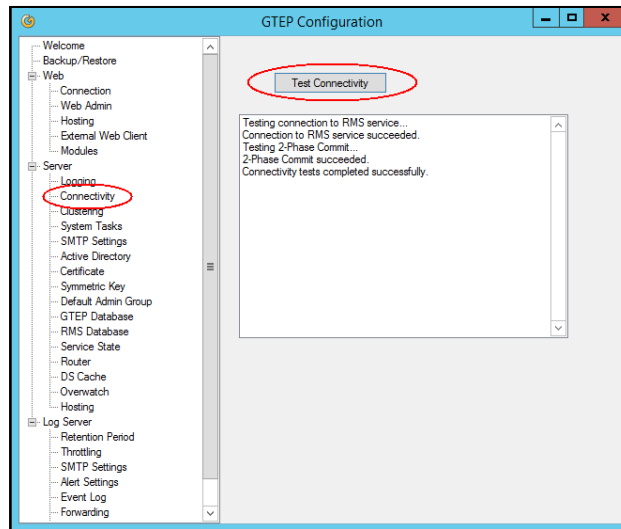
Post Installation Configuration for GigaCloud

Verify the GTEP installation:

1. In the GTEP Configuration Tool, in the tree-view in the left-hand pane, select **Server** ➤ **Connectivity**. This displays the Test Connectivity dialog.

Click **Test Connectivity**.

Information regarding the state of the GTEP installation is displayed.

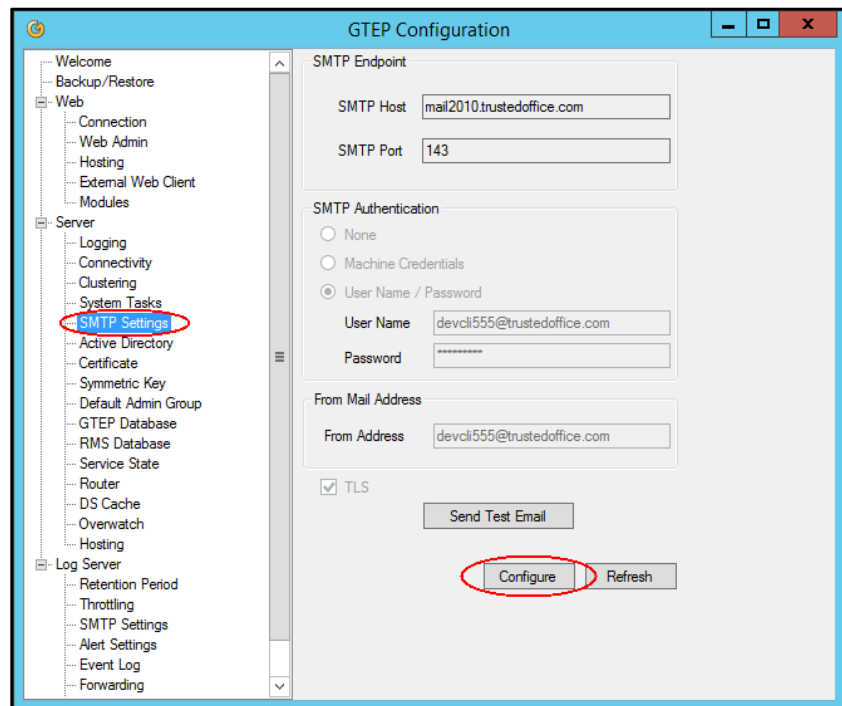


The error "MSDTC partially configured" is acceptable for single instance configuration.

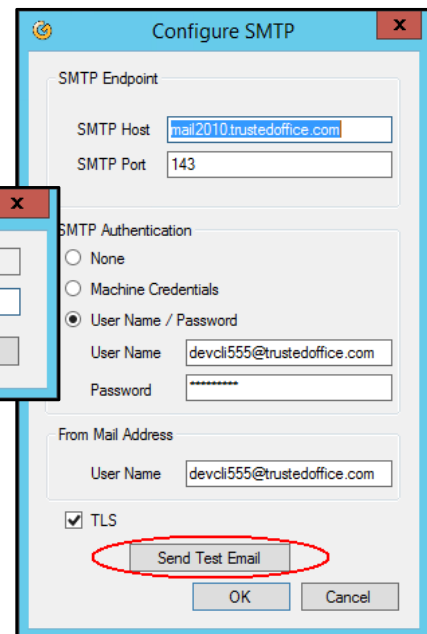
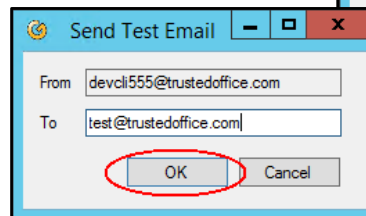
Keep the **GTEP Configuration Tool** open for the next step. ➡

Configure SMTP Settings for Invitations and Notifications:

- a. Run the **Start > All Programs > GigaTrust > GTEP > GTEP Configuration Tool**.
- b. In the tree-view in the left-hand pane, select **Server > SMTP Settings** and then click **Configure**. This displays the **Configure SMTP** Form dialog.



- c. Enter valid SMTP information in the dialog and then click **Send Test Email**. This displays the Send Test Email dialog.
- d. Enter valid To and From email addresses for the test message and then click **OK**.
- e. Verify that the test message was received.



Configure Other GTEP Settings:

All of the following GTEP settings are configured in **GTEP Administration** ➤ **Application Settings**.

a. Set the polling intervals for clients to check for updated content policies

The two polling interval settings are:

Policy Polling Interval (seconds)

Policy Polling Interval on Failure (seconds)

Policy Polling Interval is the interval, in seconds, at which the GigaTrust Desktop Client policy agent checks the server for new and updated policies.

Policy Polling Interval on Failure is the interval, in seconds, at which the Client will attempt to connect to the server for new and updated policies after a failed attempt. (A failed attempt could be due to the maximum number of concurrent policy updates already in progress.)

Both Client polling intervals are pushed down to the policy agent on the Client.

The default value for each polling setting is 120 seconds.

There are no set rules for setting the polling intervals; it is up to the organization to decide how frequently they want their clients to check for updated policies. An organization with a large user community may want to consider increasing the interval as well as adjusting the Maximum Concurrent Policy Updates setting.



After demonstrating that policy distribution is working; it is recommended that these values be set to at least 3600 or 7200 seconds.

b. Configure Alert Email Subject

The Alert sender's email address is set in the Configuration Tool under **Log Server > SMTP Settings**. If you haven't already done so, you will need to set this up too. This email address is the address from which all GTEP alerts will be sent.

The **Alert Email Subject** is defaulted to "GTEP Alerts." You can customize it if required.

c. Configure GTEP Server Address

The GTEP Server Address is defined via:

GTEP Server Address: This is the fully qualified address of the GTEP server that generates alerts. (Example: <https://adrms.trustedoffice.com/GTEP>) A link to this server is included in the alert emails. On installation this defaults to **LocalHost**.



CONFIGURATION COMPLETE!

This completes the Post Configuration Settings. Additional configuration is not necessary for a Single Instance installation.

You are now ready to use the GigaTrust GTEP Administrator Web Interface.

Go to <https://gtepserver.yourdomain.com/GTEP>

Need help setting up your organization? Use our [GigaCloud Cloud Configuration Guide](#).

Upgrading GigaCloud

On initial installation of GTEP, a backup file is created in “C:\ProgramData” and all GTEP settings are stored there as well as in their default locations. Since GTEP does not support over-installing for upgrades, and the current software must be uninstalled first, the backup is the method used to restore GTEP settings after reinstalling the software.

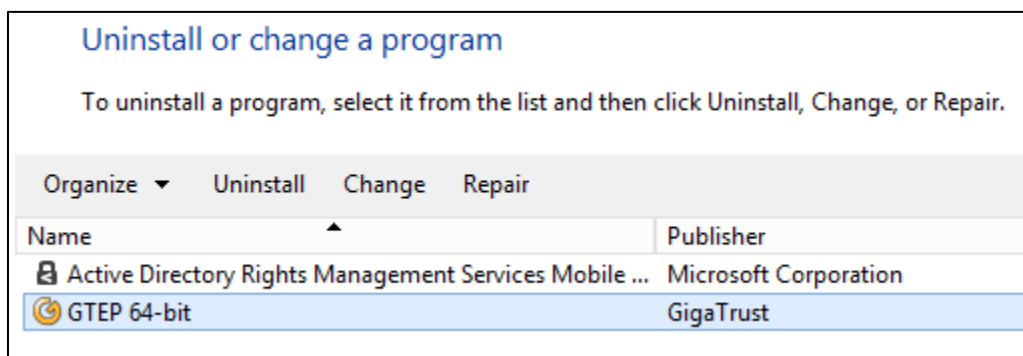


GTEP and GigaTrust Mobility Coexisting on the Same AD RMS Server

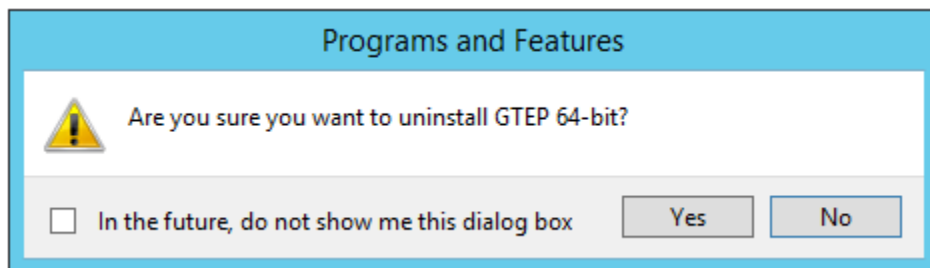
If GTEP and GigaTrust Mobility are installed on the same AD RMS server, when upgrading either software version, both must be uninstalled together and then reinstalled separately again.

Basic upgrading starts with uninstalling the current GTEP version on each instance or server it is installed on. Follow the steps below for upgrading GTEP:

1. On the first instance, or the solo GTEP instance if not in a multi-instance environment, go to **Control Panel > Programs and Features**



2. Select **GTEP 64-bit** and choose **Uninstall**.



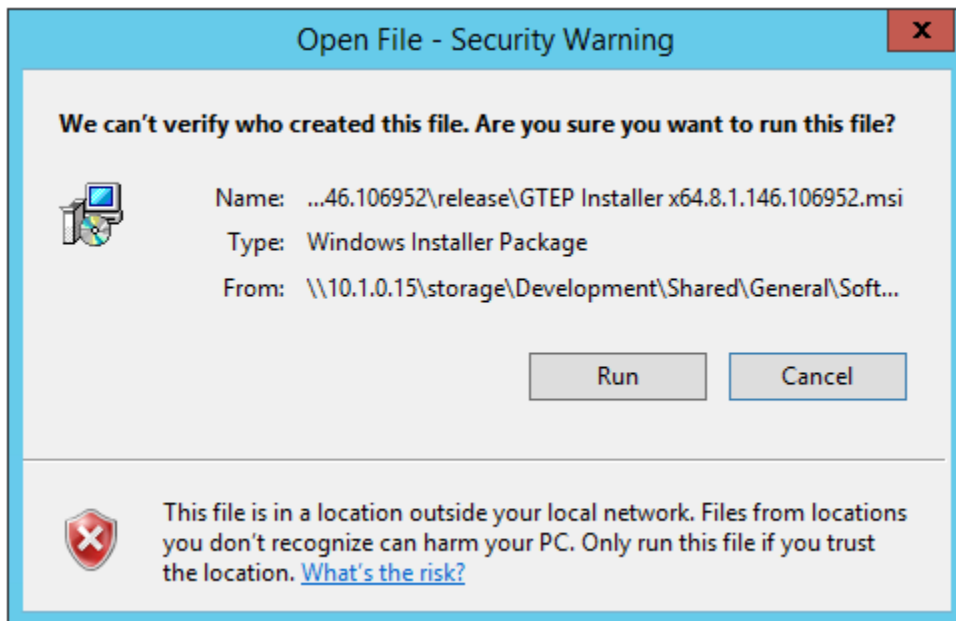
3. When the **Programs and Features > Uninstall Confirmation** prompt displays, click **Yes**.
4. A window will display with the following text: “**Please wait while Windows configures GTEP 64-bit**”
5. The window will update with the estimated time left to uninstall GTEP. When the window closes, **GTEP 64-bit** should no longer be listed in the **Control Panel**. You may have to right-click and select **Refresh** in the **Control Panel** window. If **GTEP 64-bit** is no longer listed, it has successfully uninstalled.



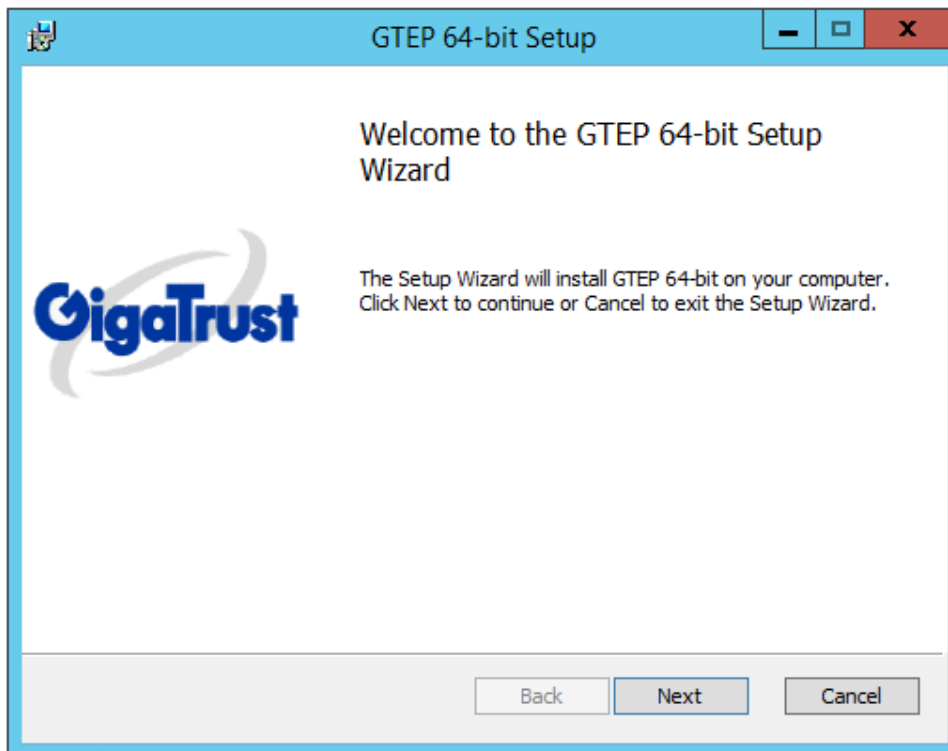
If you have multiple instances of GTEP in a clustered, or other environment, uninstall from ALL instances completely before continuing with the next step.

6. On the server instance, or the solo/single instance where GTEP was installed, navigate to the installation .MSI file. Double-click to start installation.

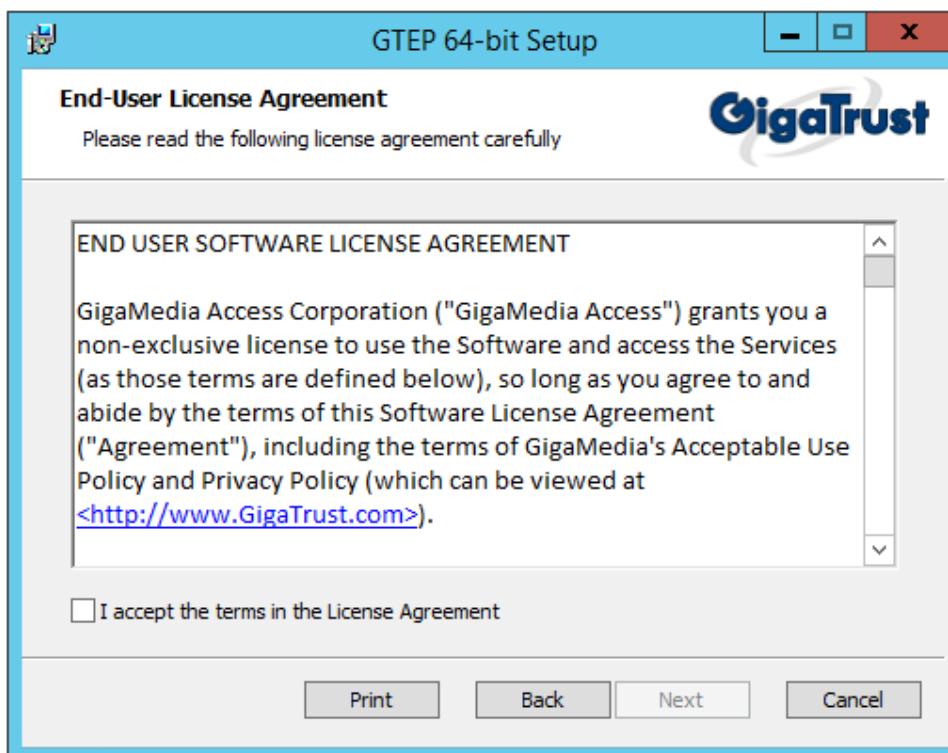
Name	Date modified	Type	Size
 GTEP Installer x64.8.1.146.106952.msi	9/18/2015 1:57 AM	Windows Installer ...	19,628 KB



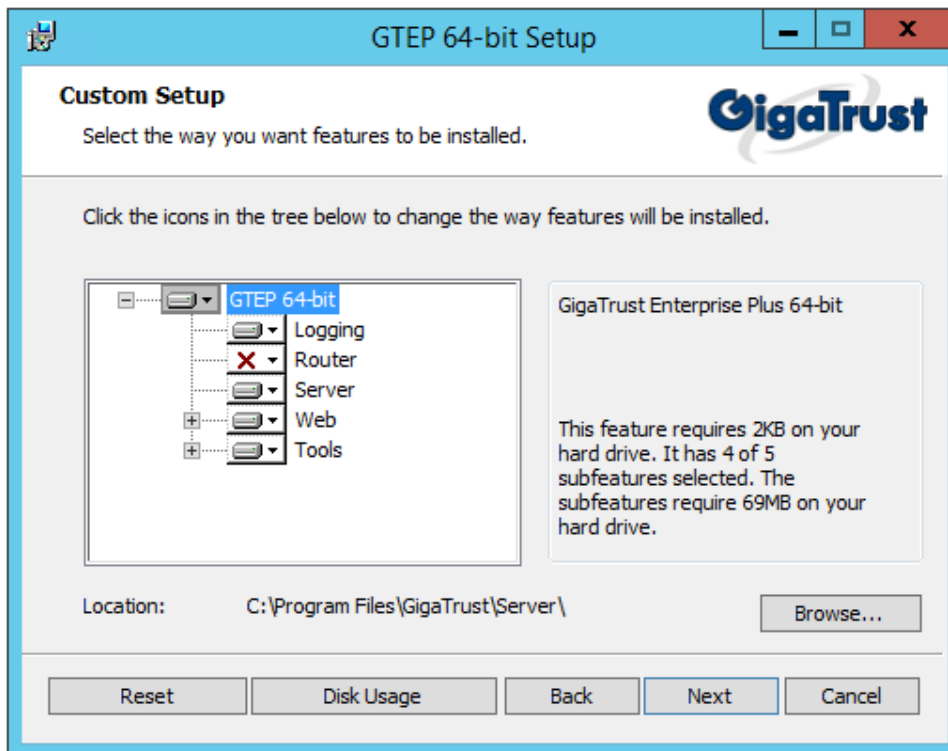
7. The **Open File – Security Warning** window appears. Select **Run**.



8. Wait for the **GTEP 64-bit Setup** window to fully load. (The **Next** button will become active.) Select **Next**.

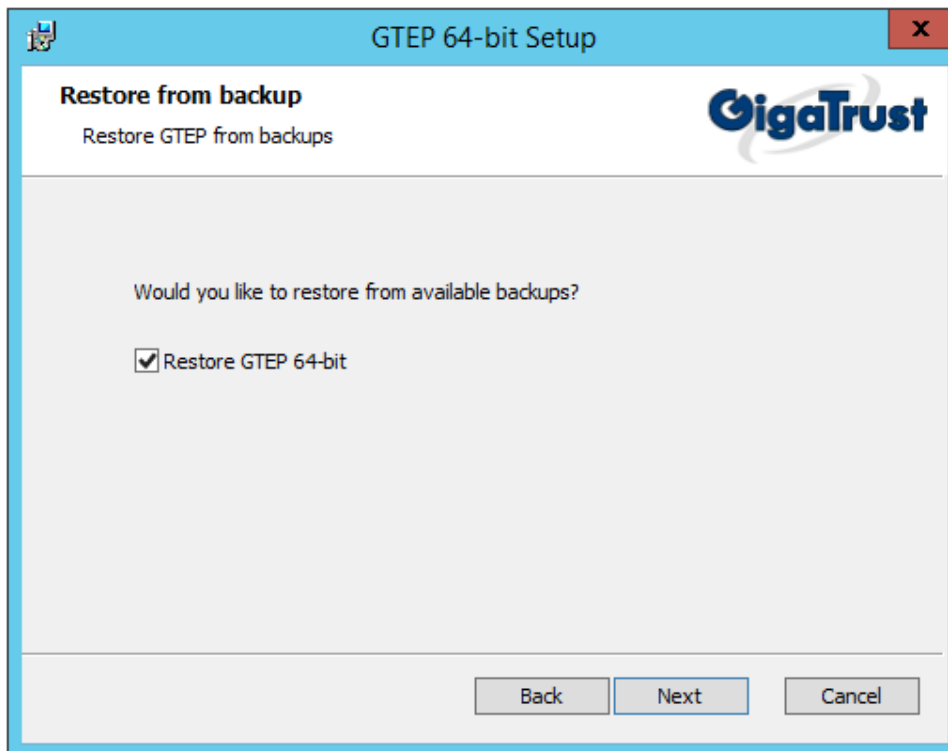


9. Check the box next to: **I accept the terms in the License Agreement** and click **Next**.

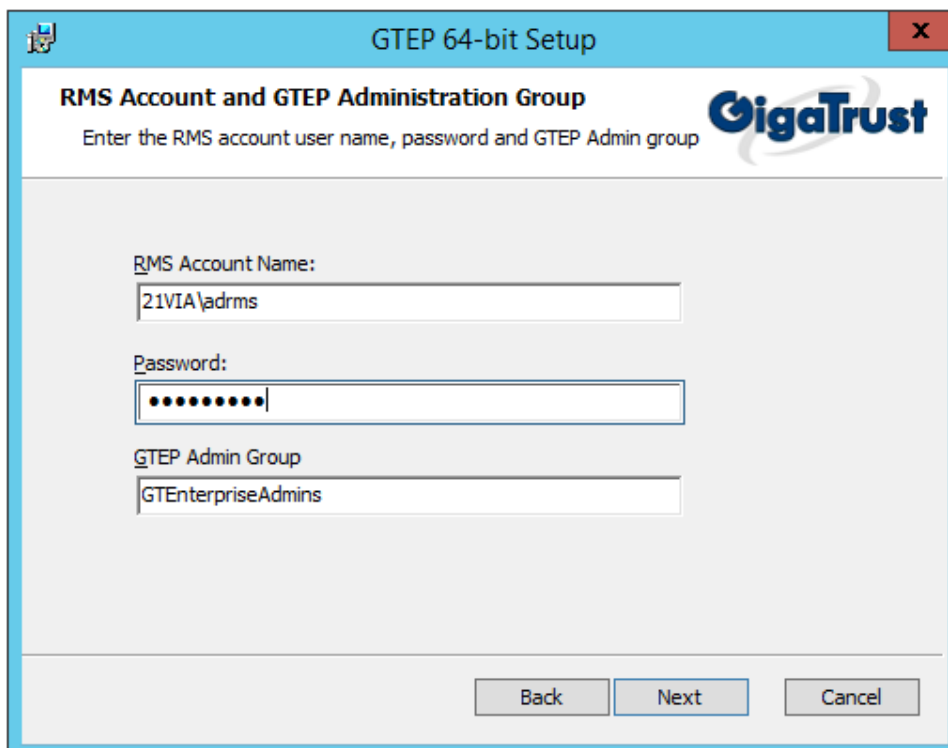


10. On the next window, if you are installing from Back Up, all of the previously installed modules will be pre-selected for reinstallation automatically. Otherwise, the default modules will be selected. Unless you need to add a feature, do not modify. You can, if needed, change the server drive where GTEP will be installed. The default location is C drive. Select Browse and choose the new location.

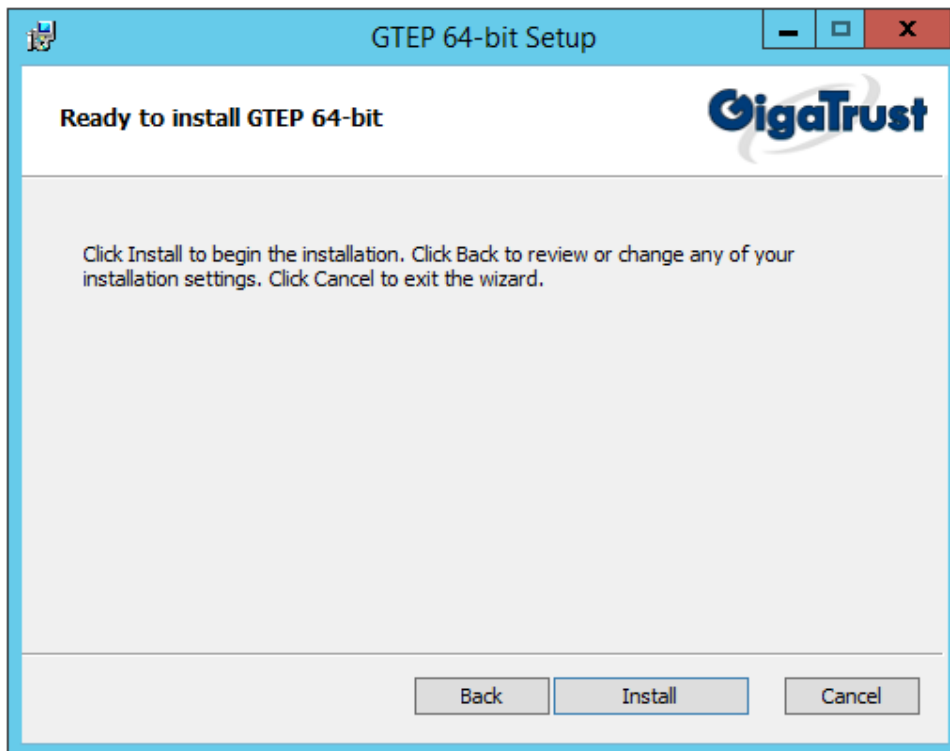
*When you are finished with custom setup, select **Next**.*



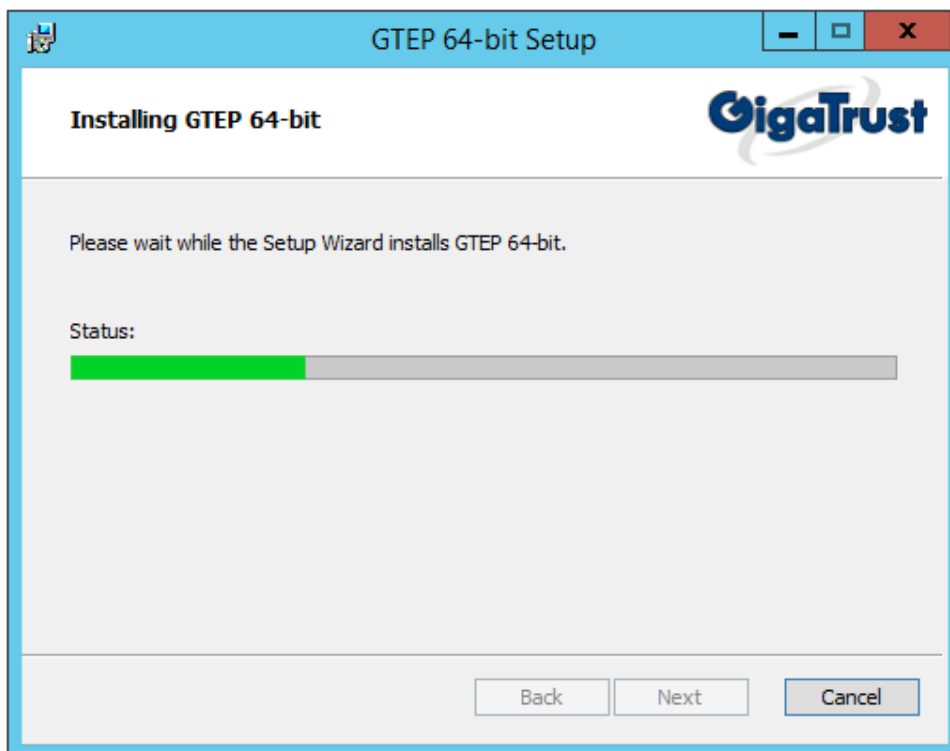
11. On the next window the **Restore GTEP 64-bit** will be pre-checked. **DO NOT UNCHECK!** Doing so will not upgrade GTEP, but will reset all settings to the factory defaults. Select **Next**.



12. Enter your current password for AD RMS from initial installation. Click **Next**.



13. On the next window, click on **Install**.



14. If prompted to upgrade the database, confirm the changes.

15. When installation is complete, close the Installer.

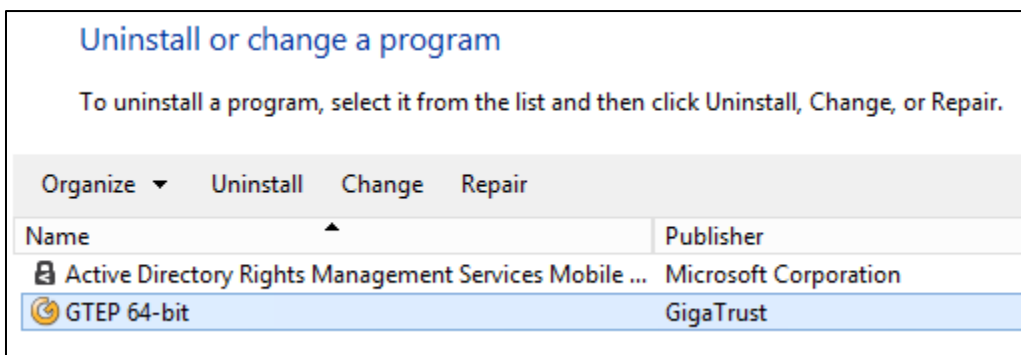
Note: *If you have a multiple GTEP instance environment, go to the next instance and repeat steps **6 – 15** until GTEP has been reinstalled on all servers.*



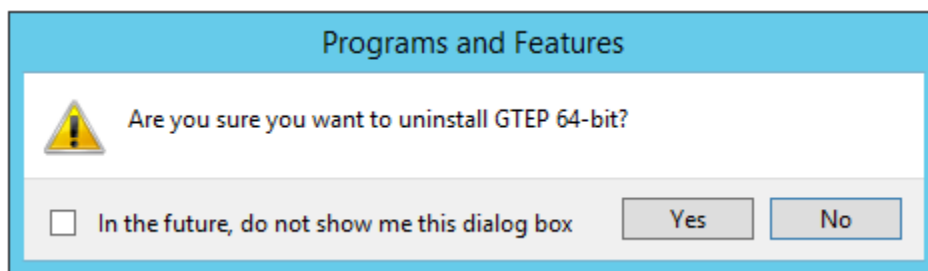
You have finished Upgrading GigaCloud.

Removing GigaCloud

1. On the first instance, or the solo GTEP instance if not in a multi-instance environment, go to **Control Panel > Programs and Features**



2. Select **GTEP 64-bit** and choose **Uninstall**.



3. When the **Programs and Features > Uninstall Confirmation** prompt displays, click **Yes**.
4. A window will display with the following text: "**Please wait while Windows configures GTEP 64-bit**"
5. The window will update with the estimated time left to uninstall GTEP. When the window closes, **GTEP 64-bit** should no longer be listed. You may have to right-click and select **Refresh** in the **Control Panel** window. If **GTEP 64-bit** is no longer listed, it has successfully uninstalled.



If you have multiple instances of GTEP in a clustered, or other environment, continue by uninstalling from ALL instances.

6. If you are not reinstalling GTEP, or upgrading it, you can optionally remove the GTEP SQL Server database.

7. After removal, the following references to GTEP will remain on the AD RMS/SQL servers which you may remove if necessary:
 - a. On the AD RMS server(s) the **ProgramData** folder will contain a GigaTrust folder which stores the backup files for reinstallation.
 - b. On the AD RMS server(s) the following file path will still exist: C:\Program Files\GigaTrust\Server\GTEP\Web\GTEPEXT\software and any GTDC software installation files will still reside here.
 - c. On the AD RMS server(s) the following file path will still exist:
C:\Users\[Administrator]\AppData\Local\GigaMedia_Access_Corporat\LogClient.exe_Url_[Random String]\9.0.0.0.

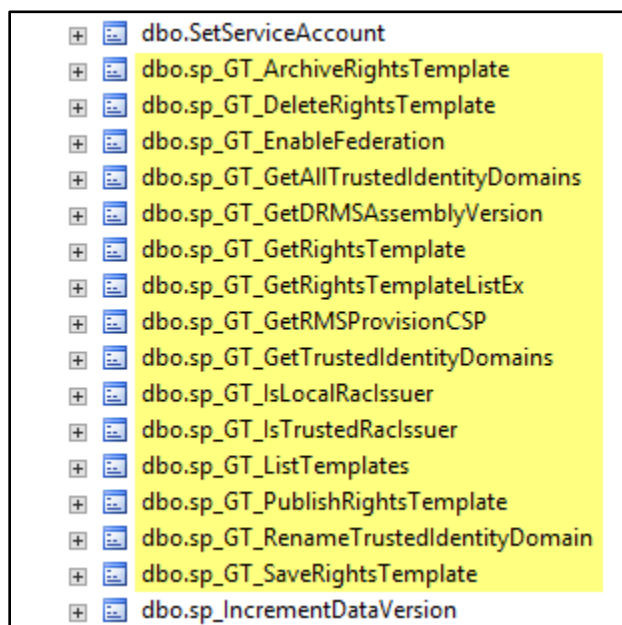
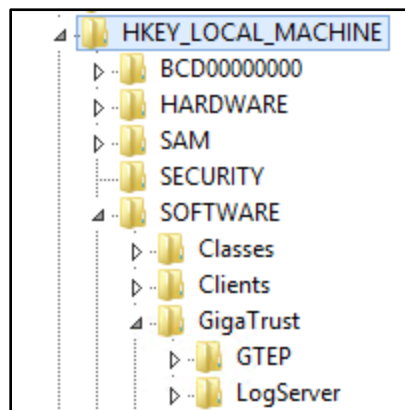
This folder contains the user.config file.

- d. The Registry will still contain the following path with GTEP settings:

[HKEY_LOCAL_MACHINE\SOFTWARE\GigaTrust\]

See right screen shot.

- e. The GTEP database will still reside on your SQL server/cluster.
- f. SQL Stored Procedures will still be contained in the [DRMS_Config_<Your DB Info>] under Programmability > Stored Procedures. The procedure names will be prefixed with “sp_GT_” – see screen shot below.



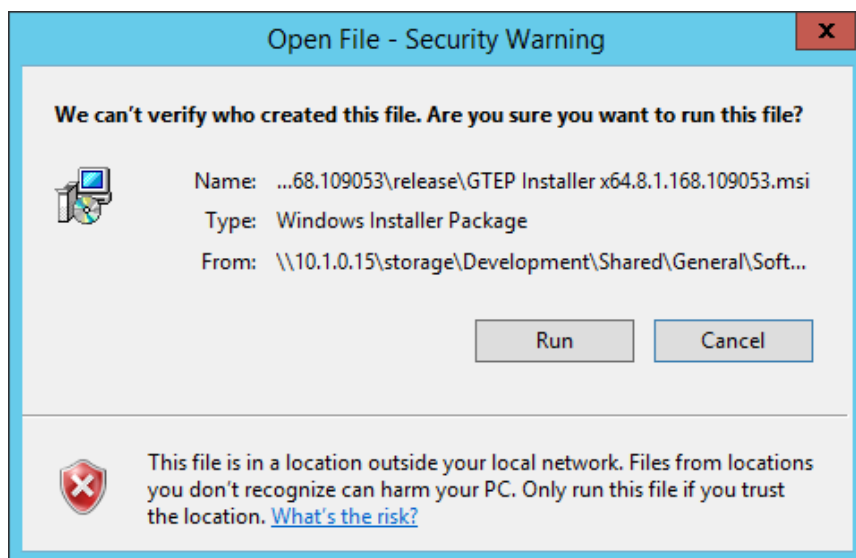
You have completed the removal of GTEP.

Appendix A: Separate GTEP Log Client Installation

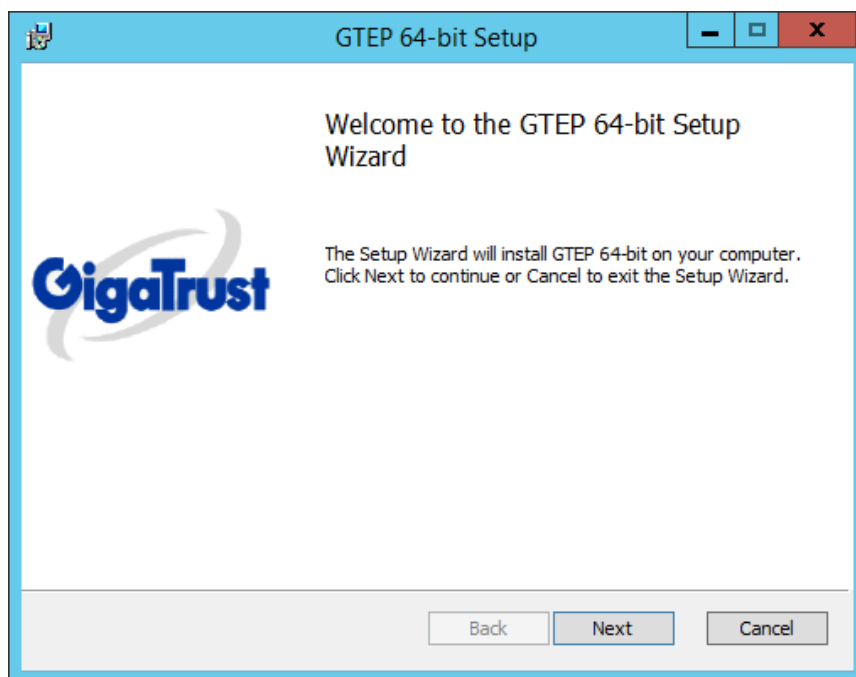
The GTEP Log Client may be installed on another server in your cloud, Enterprise or from a desktop computer for easier accessibility to monitor your AD RMS services through GTEP. The Log Client can be installed on Windows Server 2008/2012 R2, 2016 or Windows 7/8/10 64 bit. .NET 4.6.2 is a required prerequisite. To install on another server or on a desktop image, follow the steps below.

1. Start the Installation:

Run **GTEP64VERSION.MSI**.



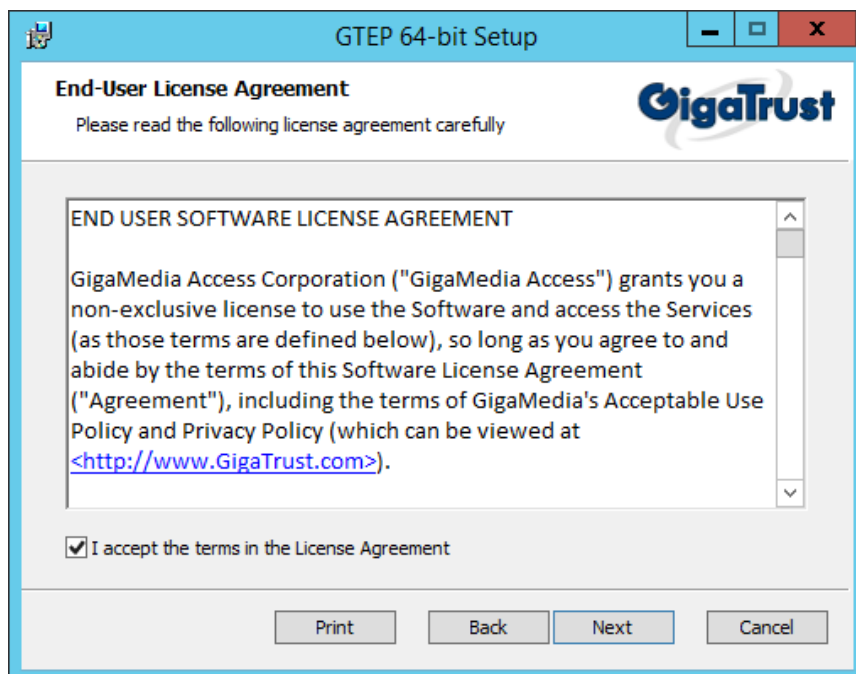
2. The Welcome screen appears. Click **Next** to continue.



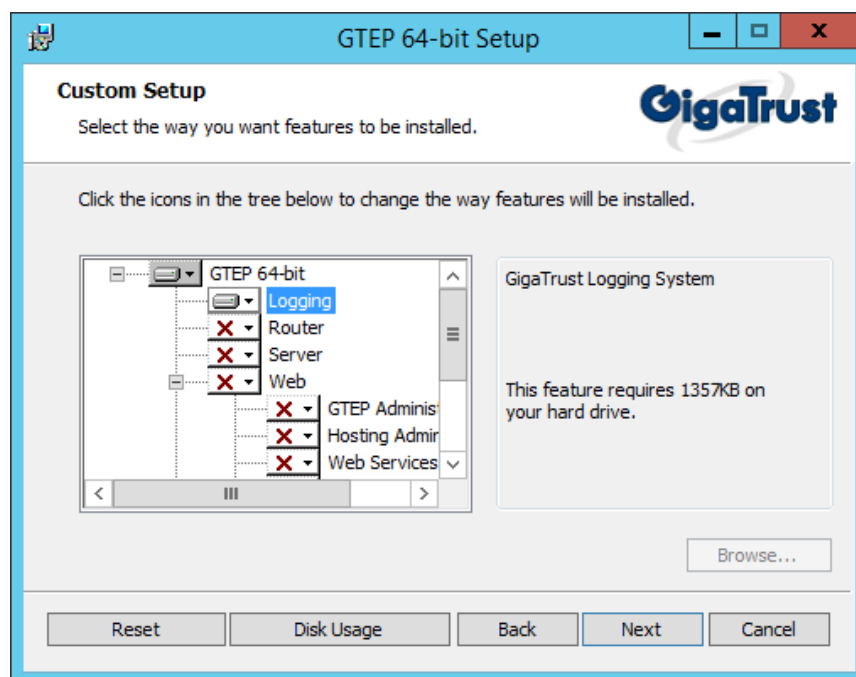
3. End-User License Agreement:

Read and accept the **GigaTrust End-User License Agreement**.

Click **Next** to continue.

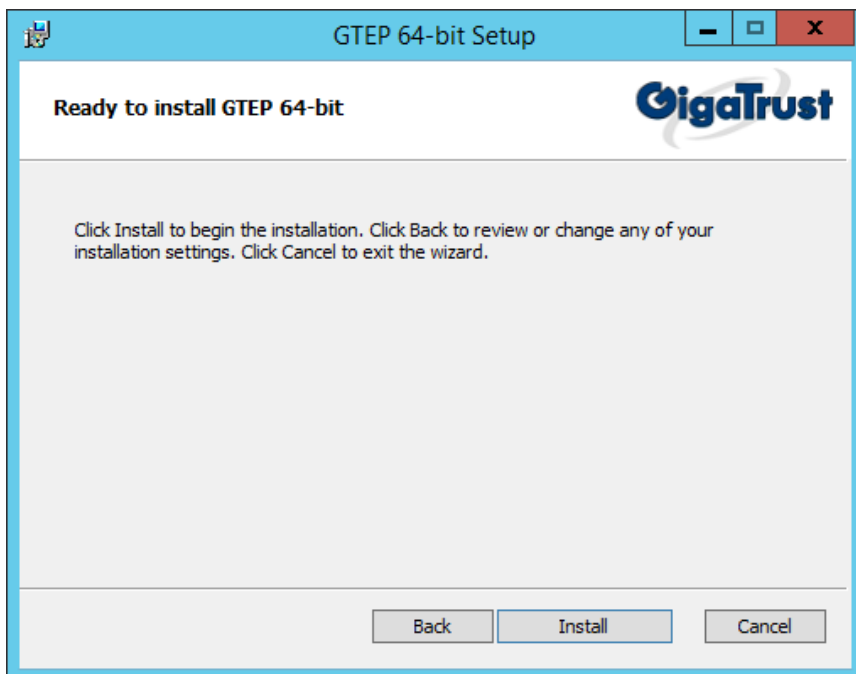


4. Select the **Logging** component only. Deselect ALL others.

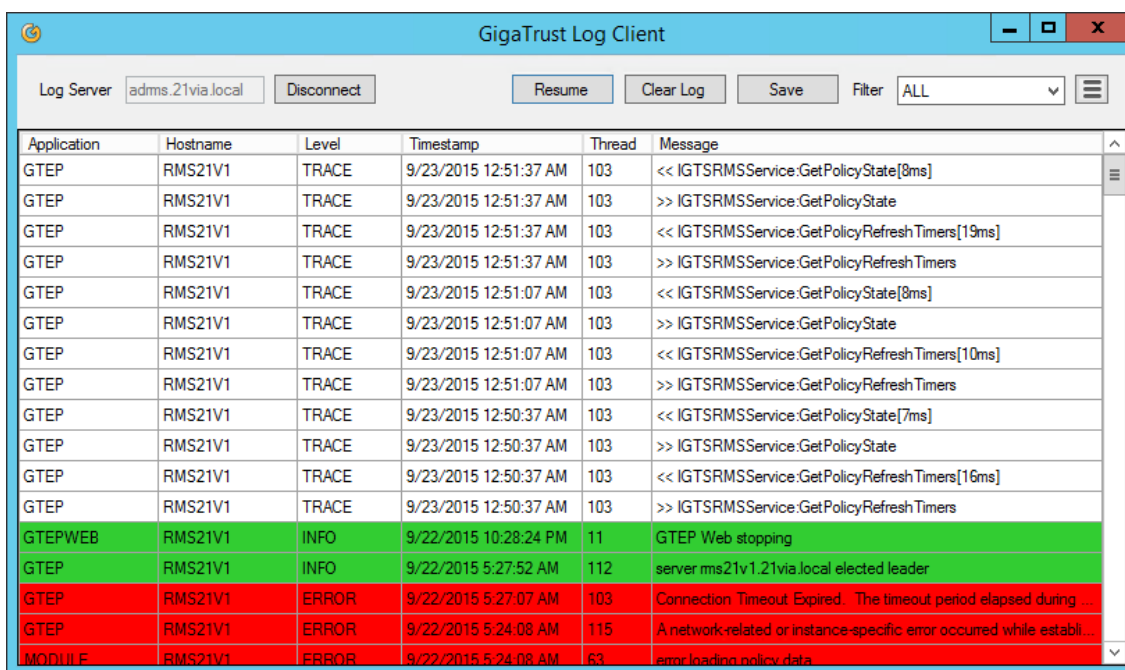


- Click **Next** through additional steps and then click **Install** on the “Ready to install GTEP 64-bit” window.

Note: There may be a delay, please click only once.



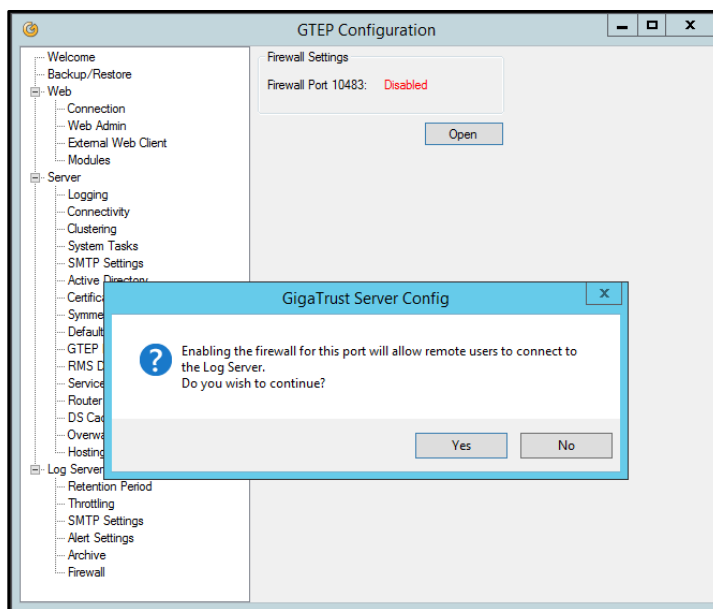
- When installation has finished go to **Programs** ➤ **GigaTrust** and select **GigaTrust Log Client**. In **Log Server** enter “localhost” or the FQDN for the GTEP instance (i.e. ADRMS.YourDomain.com) and select **Connect**. For more detailed Log Client instructions use our [GTEP Troubleshooting Guide](#).





Firewall Permissions

The AD RMS firewall port #10483 can be opened to allow the Log Client to connect to the server. This can be accomplished in the GTEP Configuration Tool. Go to the **Firewall** node and select **Open** and confirm the change.



Appendix B: Miscellaneous Information

Enterprise Templates

All work with Enterprise-level templates must be performed through GTEP, not through the AD RMS Administrator. This ensures proper enterprise template functionality.

Software Version

Desktop Clients v11.3 and higher are supported.

GTEP Service Account – Language/Locale must be English

GTEP service user account's language/locale must be set to English. Otherwise, some reports may hang when being generated.

1. Log on to the GTEP server as the GTEP service account.
2. If necessary, change the language and locale to English.
3. Stop the GTEP services.
4. Restart IIS.
5. Start the GTEP services.

Appendix C: GTEP Service Account

If the administrator does not have access to the AD RMS Service account, the following is required before installation of GigaCloud Administration. Instead of supplying the AD RMS Server account credentials during installation, the administrator will supply credentials for a GTEP Service account for which the administrator does have full access.

1. Create a user in Active Directory called **GTEPService**.
2. In Active Directory, add the GTEPService account to the **GTEnterpriseAdmins** group. (See [Prerequisites](#).)
3. On the AD RMS server(s), add the GTEPService account to the Local Group **AD RMS Service Group**.
4. On the AD RMS SQL Server, add GTEPService to the **SQL SYSADMIN** group.
Verify that the GTEPService account has the same permissions as the RMSService account.
5. Confirm prerequisites are fulfilled. Then install GigaCloud Administration.

Note: When asked to provide the RMSService User Name and Password, enter the GTEPService User Name and Password instead.

Appendix D: Troubleshooting

Start by connecting to the GigaTrust Log Client. This will display real-time errors to assist in troubleshooting GTEP. For further troubleshooting assistance, use the [GigaCloud Troubleshooting Guide](#).